Module 13

Section 8.4 | Vectors

Math 130: Advanced Technical Mathematics | Unit 3

Section 8.4

Vectors

Section 8.4 — Topics Covered (5 Skills)

- Writing a vector in component form given initial and terminal points
- Vector addition and scalar multiplication: component form
- Finding magnitude and direction of a vector given its graph
- Finding the direction angle of a vector in ai + bj form
- Finding the components of a vector given its graph

Vectors and Scalars

- Scalar: a quantity with magnitude only
- Examples: speed, temperature, mass, time
- Represented by a single number
- Vector: a quantity with both magnitude AND direction
- Examples: velocity, force, displacement, wind
- Represented by a directed line segment (arrow)

Notation

- Vectors are written as: v, or with an arrow above
- Bold letters in textbooks: v
- Two-point notation: vector from A to B
- Component form: v = <a, b> or v = ai + bj
- Magnitude: |v| or ||v||

Vector in Component Form

Component Form: Given initial point $P(x_1, y_1)$ and terminal point $Q(x_2, y_2)$, the vector PQ in component form is $<x_2 - x_1, y_2 - y_1>$.

- The components represent horizontal and vertical displacement
- Component form is independent of position (translation invariant)
- Two vectors are equal if and only if they have the same components
- Example: $P(1, 3)$ to $Q(4, 7)$ → $v = <4-1, 7-3> = <3, 4>$

Example — Component Form from Points

Find the component form of the vector with initial point A(−2, 5) and terminal point B(3, −1).

- $v = <x_2 - x_1, y_2 - y_1>$
- $v = <3 - (-2), -1 - 5>$
- $v = <5, -6>$
- This means: 5 units right and 6 units down

Finding Components of a Vector from Its Graph

- When a vector is shown on a coordinate grid:
- Step 1: Identify the initial point (tail) coordinates
- Step 2: Identify the terminal point (tip/head) coordinates
- Step 3: Subtract: component = <x_tip − x_tail, y_tip − y_tail>
- Alternatively, if given magnitude $|v|$ and direction angle θ:
- Horizontal component: $v_x = |v| \cos \theta$
- Vertical component: $v_y = |v| \sin \theta$
- So $v = \langle |v| \cos \theta, |v| \sin \theta \rangle$

Example — Components from a Graph

A vector starts at (1, 2) and ends at (−3, 5) on a graph. Find its components.

- $v = \langle -3 - 1, 5 - 2 \rangle = \langle -4, 3 \rangle$
- Magnitude: $|v| = \sqrt{16 + 9} = \sqrt{25} = 5$
- Direction: $\theta = \tan^{-1}(3 / -4) \approx 143.1°$ (in Q II since $x < 0$, $y > 0$)

Vector Addition and Scalar Multiplication

Addition: <a, b> + <c, d> = <a + c, b + d>

Subtraction: <a, b> − <c, d> = <a − c, b − d>

Scalar multiplication: k<a, b> = <ka, kb>

In i, j form: (ai + bj) + (ci + dj) = (a+c)i + (b+d)j

Geometric meaning: Vector addition is 'tip-to-tail' placement (polygon method) or the diagonal of a parallelogram (parallelogram method).

Example — Vector Addition and Scalar Multiplication

Given $u = \langle 3, -2 \rangle$ and $v = \langle -1, 5 \rangle$. Find: (a) $u + v$, (b) $3u - 2v$.

- (a) $u + v = \langle 3 + (-1), -2 + 5 \rangle = \langle 2, 3 \rangle$
- (b) $3u = \langle 9, -6 \rangle$, $2v = \langle -2, 10 \rangle$
- $3u - 2v = \langle 9 - (-2), -6 - 10 \rangle = \langle 11, -16 \rangle$
- Verify magnitude of (a): $|u + v| = \sqrt{4 + 9} = \sqrt{13} \approx 3.61$

Magnitude and Direction of a Vector

Magnitude: $|v| = \sqrt{a^2 + b^2}$ for $v = \langle a, b \rangle$

Direction angle: $\theta = \tan^{-1}(b / a)$ (adjust for quadrant)

Q I: $\theta = \tan^{-1}(b/a)$

Q II or III: $\theta = \tan^{-1}(b/a) + 180°$

Q IV: $\theta = \tan^{-1}(b/a) + 360°$ (to get positive angle)

Example — Magnitude and Direction from Graph

A vector on a graph goes from the origin to the point (−3, 4). Find its magnitude and direction angle.

- $v = \langle -3, 4 \rangle$
- Magnitude: $|v| = \sqrt{9 + 16} = \sqrt{25} = 5$
- $\tan^{-1}(4 / -3) = \tan^{-1}(-1.333) \approx -53.1°$
- Since the point is in Q II, add 180°: $\theta = -53.1° + 180° = 126.9°$
- The vector has magnitude 5 and direction ≈ 126.9°

Example — Direction Angle of ai + bj

Find the direction angle of $v = -5i + 12j$.

- $v = \langle -5, 12 \rangle \rightarrow a = -5, b = 12$
- $|v| = \sqrt{(25 + 144)} = \sqrt{169} = 13$
- $\tan^{-1}(12 / -5) = \tan^{-1}(-2.4) \approx -67.4°$
- Since $a < 0$ and $b > 0$ (Q II): $\theta = -67.4° + 180° = 112.6°$
- Direction angle is approximately 112.6°

Example — Direction Angle (Q III and Q IV)

Find the direction angles of: (a) $v = -3i - 7j$, (b) $w = 4i - 2j$.

- (a) $v = \langle -3, -7 \rangle$: $\tan^{-1}(-7/-3) = \tan^{-1}(2.333) \approx 66.8°$
- Q III (both negative): $\theta = 66.8° + 180° = 246.8°$
- (b) $w = \langle 4, -2 \rangle$: $\tan^{-1}(-2/4) = \tan^{-1}(-0.5) \approx -26.6°$
- Q IV ($a > 0$, $b < 0$): $\theta = -26.6° + 360° = 333.4°$

Unit Vectors

Unit Vector: A vector with magnitude 1. The unit vector in the direction of v is: $\hat{u} = v / |v| = \langle a/|v|, b/|v| \rangle$.

- Standard unit vectors: $i = \langle 1, 0 \rangle$ and $j = \langle 0, 1 \rangle$
- Any vector $v = \langle a, b \rangle$ can be written as $v = ai + bj$
- To find a unit vector in the direction of v, divide v by its magnitude
- Example: $v = \langle 3, 4 \rangle$, $|v| = 5 \rightarrow \hat{u} = \langle 3/5, 4/5 \rangle = \langle 0.6, 0.8 \rangle$

Vector Addition: Geometric Methods

- Polygon Method (Tip-to-Tail):
- Place the tail of the second vector at the tip of the first
- The resultant goes from the tail of the first to the tip of the last
- Works for adding any number of vectors
- A + B + C: chain them tip-to-tail

Parallelogram Method

- Place both vectors at the same initial point
- Complete the parallelogram
- The diagonal from the shared point is the resultant
- Best for adding exactly two vectors
- Note: order doesn't matter (A + B = B + A)

Example — Boating Across a River

A boat crosses a 0.4-mile wide river at 8 mph. The current is 6 mph. What is the resultant speed and direction?

- Boat velocity: $v_{boat} = \langle 0, 8 \rangle$ (perpendicular to current, heading across)
- Current velocity: $v_{current} = \langle 6, 0 \rangle$ (downstream)
- Resultant: $v = \langle 6, 8 \rangle$
- Speed = $|v| = \sqrt{36 + 64} = \sqrt{100} = 10$ mph
- Direction: $\theta = \tan^{-1}(8/6) \approx 53.1°$ from the downstream direction

Example — Wind Components for Takeoff

Departing runway 14 with winds from 080° at 18 knots. Find the headwind and crosswind components.

- Runway 14 = heading 140°. Wind from 080° → angle between = 60° off nose
- Headwind component = $18 \cos 60° = 18 \times 0.5 = 9$ knots
- Crosswind component = $18 \sin 60° = 18 \times 0.866 \approx 15.6$ knots
- Verify: $\sqrt{(9^2 + 15.6^2)} = \sqrt{(81 + 243)} = \sqrt{324} = 18$ ✓
- This is a quartering headwind with a strong crosswind component

Example — Lift in a Banked Turn

To maintain level flight in a turn, the vertical lift component must equal the aircraft weight. If the bank angle is 60°, what total lift is needed for a 3,200 lb aircraft?

- Vertical component of lift = $L \cos \theta$ must equal weight W
- $L \cos 60° = 3{,}200$ lb
- $L = 3{,}200 / \cos 60° = 3{,}200 / 0.5 = 6{,}400$ lb
- The aircraft needs twice its weight in total lift at 60° bank
- At 30°: $L = 3{,}200 / \cos 30° \approx 3{,}695$ lb
- At 80°: $L = 3{,}200 / \cos 80° \approx 18{,}427$ lb (extreme!)

Example — Adding Vectors Given Magnitude and Direction

Add: v (magnitude 17, direction 33°) and u (magnitude 24, direction 95°).

- $v = \langle 17 \cos 33°, 17 \sin 33° \rangle = \langle 14.26, 9.26 \rangle$
- $u = \langle 24 \cos 95°, 24 \sin 95° \rangle = \langle -2.09, 23.91 \rangle$
- $v + u = \langle 14.26 + (-2.09), 9.26 + 23.91 \rangle = \langle 12.17, 33.17 \rangle$
- Magnitude $= \sqrt{(148.1 + 1100.2)} = \sqrt{1248.3} \approx 35.3$
- Direction $= \tan^{-1}(33.17 / 12.17) \approx 69.8°$ (Q I)

Equilibrium of Forces

- An object is in equilibrium when the net force is zero
- This means: the sum of all horizontal components = 0
- and the sum of all vertical components = 0
- For any force F at angle θ:
- Horizontal: $F_x = F \cos \theta$
- Vertical: $F_y = F \sin \theta$
- Common applications: rope tension, inclined planes, suspension systems
- Strategy: set up equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$, then solve

Example — Equilibrium: Rope Tension

A 180-lb climber is held by a rope at 60° from horizontal, pushing against the cliff with force F. Find the rope tension T and force F.

- Equilibrium: sum of forces = 0 in both directions
- Vertical: $T \sin 60° = 180 \rightarrow T = 180 / \sin 60° = 180 / 0.866 \approx 207.8$ lb
- Horizontal: $F = T \cos 60° = 207.8 \times 0.5 \approx 103.9$ lb
- The rope tension is about 208 lb; the climber pushes with about 104 lb

Example — Ship Navigation with Current

A ship moves at 17.5 mph heading 26.3° N of E, but actually moves 19.3 mph at 33.7° N of E. Find the current velocity.

- Ship velocity: $v_s = <17.5 \cos 26.3°, 17.5 \sin 26.3°> = <15.68, 7.76>$
- Actual velocity: $v_a = <19.3 \cos 33.7°, 19.3 \sin 33.7°> = <16.06, 10.71>$
- Current = $v_a - v_s = <16.06 - 15.68, 10.71 - 7.76> = <0.38, 2.95>$
- Current speed = $\sqrt{(0.14 + 8.70)} = \sqrt{8.84} \approx 2.97$ mph
- Current direction = $\tan^{-1}(2.95/0.38) \approx 82.7°$ N of E (nearly due north)

Module 13 Summary

- Vectors have both magnitude and direction (unlike scalars)
- Component form: $v = \langle a, b \rangle = ai + bj$, found from initial/terminal points
- Addition/subtraction: add/subtract corresponding components
- Scalar multiplication: multiply each component by the scalar
- Magnitude: $|v| = \sqrt{(a^2 + b^2)}$
- Direction angle: $\theta = \tan^{-1}(b/a)$, adjusted for quadrant
- Applications: wind, current, navigation, forces, equilibrium
- 5 ALEKS topics total for this module